

OpenShift 4 HAProxy Runbook

Load Balancer Configuration for 3 Masters, 2 Workers, 1 Bootstrap Node

1. Architecture Overview

This document outlines the installation and configuration of an HAProxy load balancer for a User-Provisioned Infrastructure (UPI) OpenShift 4 cluster.

This load balancer routes traffic for an OpenShift cluster configured with **schedulable master nodes** (where the ingress router runs on the masters). Worker nodes are excluded from the LB configuration as they do not host the ingress controllers in this architecture.

Node Topology

- **1x Bootstrap Node** (Required for initial cluster bootstrap)
- **3x Master Nodes** (Control Plane + Ingress components)
- **2x Worker Nodes** (Compute only, not load balanced)

Required Port Forwarding

Port	Service	Target Nodes
6443	Kubernetes API Server	Bootstrap, Masters (0,1,2)
22623	Machine Config Server (MCS)	Bootstrap, Masters (0,1,2)
80	Ingress (HTTP)	Masters (0,1,2)
443	Ingress (HTTPS)	Masters (0,1,2)

2. Prerequisites

- A dedicated VM for HAProxy running RHEL, CentOS, or Rocky Linux.
- Root or `sudo` privileges on the HAProxy VM.
- Static IP addresses assigned to the Bootstrap and Master nodes.

3. Installation & OS Configuration

3.1 Install HAProxy

Install the HAProxy package from the standard repositories:

```
sudo dnf update -y
sudo dnf install haproxy -y
```

3.2 Configure SELinux

Allow HAProxy to make outbound network connections. This is required for health checks and traffic routing:

```
sudo setsebool -P haproxy_connect_any 1
```

4. Configuration

4.1 Backup Default Configuration

```
sudo cp /etc/haproxy/haproxy.cfg /etc/haproxy/haproxy.cfg.bak
```

4.2 Apply OpenShift Routing Rules

Edit the configuration file (`sudo vi /etc/haproxy/haproxy.cfg`) and replace the contents with the following snippet. (Note: Replace the `<BOOTSTRAP_IP>` and `<MASTER_X_IP>` variables with your actual static IP addresses.)

```
global
    log          127.0.0.1 local2
    chroot       /var/lib/haproxy
    pidfile      /var/run/haproxy.pid
    maxconn      4000
    user         haproxy
    group        haproxy
    daemon
    stats socket /var/lib/haproxy/stats

defaults
    mode          tcp
    log           global
    option        tcplog
    option        dontlognull
    retries       3
    timeout http-request 10s
```

```
    timeout queue            1m
    timeout connect          10s
    timeout client           1m
    timeout server           1m
    timeout http-keep-alive  10s
    timeout check            10s
    maxconn                  3000

# -----
# OpenShift API Server
# -----
frontend openshift-api-server
    bind *:6443
    default_backend openshift-api-server-backend

backend openshift-api-server-backend
    balance source
    server bootstrap <BOOTSTRAP_IP>:6443 check
    server master0 <MASTER0_IP>:6443 check
    server master1 <MASTER1_IP>:6443 check
    server master2 <MASTER2_IP>:6443 check

# -----
# Machine Config Server (MCS)
# -----
frontend machine-config-server
    bind *:22623
    default_backend machine-config-server-backend

backend machine-config-server-backend
    balance source
    server bootstrap <BOOTSTRAP_IP>:22623 check
    server master0 <MASTER0_IP>:22623 check
    server master1 <MASTER1_IP>:22623 check
    server master2 <MASTER2_IP>:22623 check

# -----
# Ingress HTTP (Router running on Masters)
# -----
frontend ingress-http
    bind *:80
    default_backend ingress-http-backend

backend ingress-http-backend
    balance source
    server master0 <MASTER0_IP>:80 check
    server master1 <MASTER1_IP>:80 check
    server master2 <MASTER2_IP>:80 check

# -----
# Ingress HTTPS (Router running on Masters)
```

```
# -----  
frontend ingress-https  
    bind *:443  
    default_backend ingress-https-backend  
  
backend ingress-https-backend  
    balance source  
    server master0 <MASTER0_IP>:443 check  
    server master1 <MASTER1_IP>:443 check  
    server master2 <MASTER2_IP>:443 check
```

5. Service Management

5.1 Enable and Start

Enable HAProxy to start on boot and launch the service immediately:

```
sudo systemctl enable --now haproxy
```

5.2 Verify Status

Check that the service is running actively without syntax errors:

```
sudo systemctl status haproxy
```

6. Day-2 Operations (Post-Install)

Once the OpenShift cluster installation is fully complete and the Bootstrap node is safely destroyed, you must remove it from the load balancer rotation to prevent unnecessary health check failures.

1. Edit `/etc/haproxy/haproxy.cfg`.
2. Delete or comment out (`#`) the `server bootstrap` lines under both the `openshift-api-server-backend` and `machine-config-server-backend` sections.
3. Reload the HAProxy configuration without dropping active connections:

```
sudo systemctl reload haproxy
```

Important: Do not use `systemctl restart haproxy` for configuration updates, as reloading gracefully applies the changes without terminating active connections.